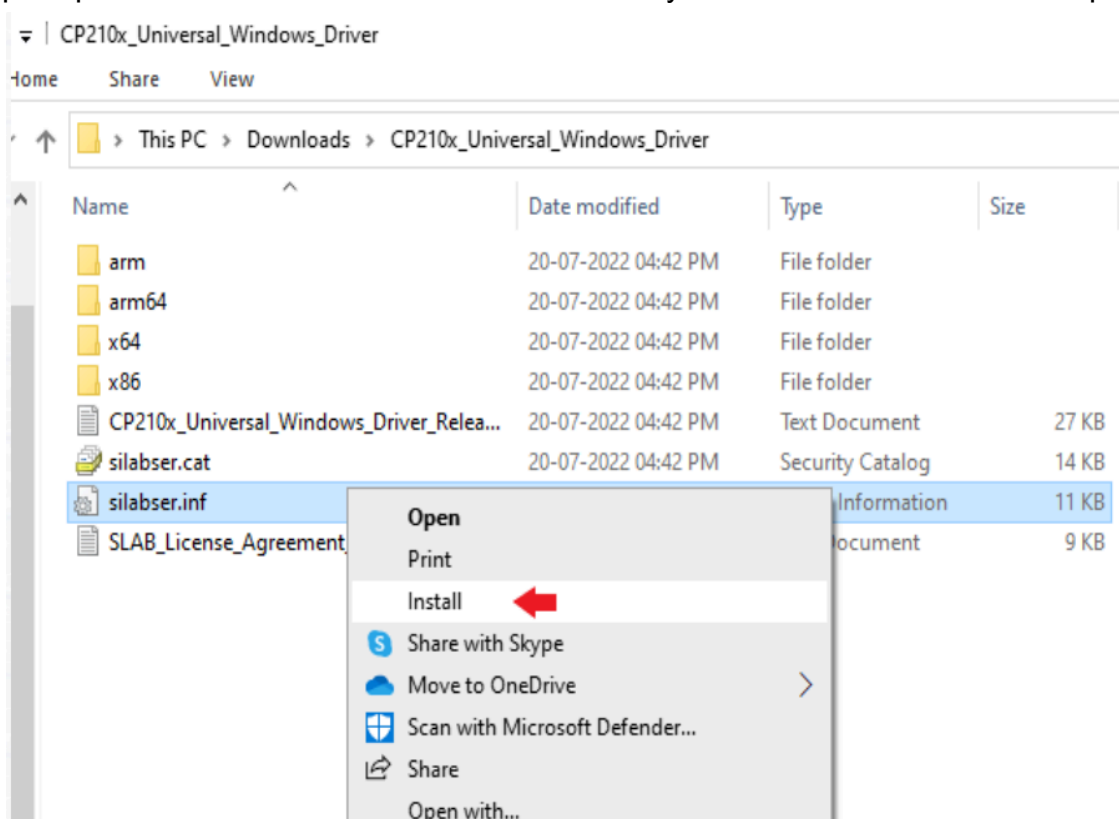


Initial Set Up for BuildClub Workshop

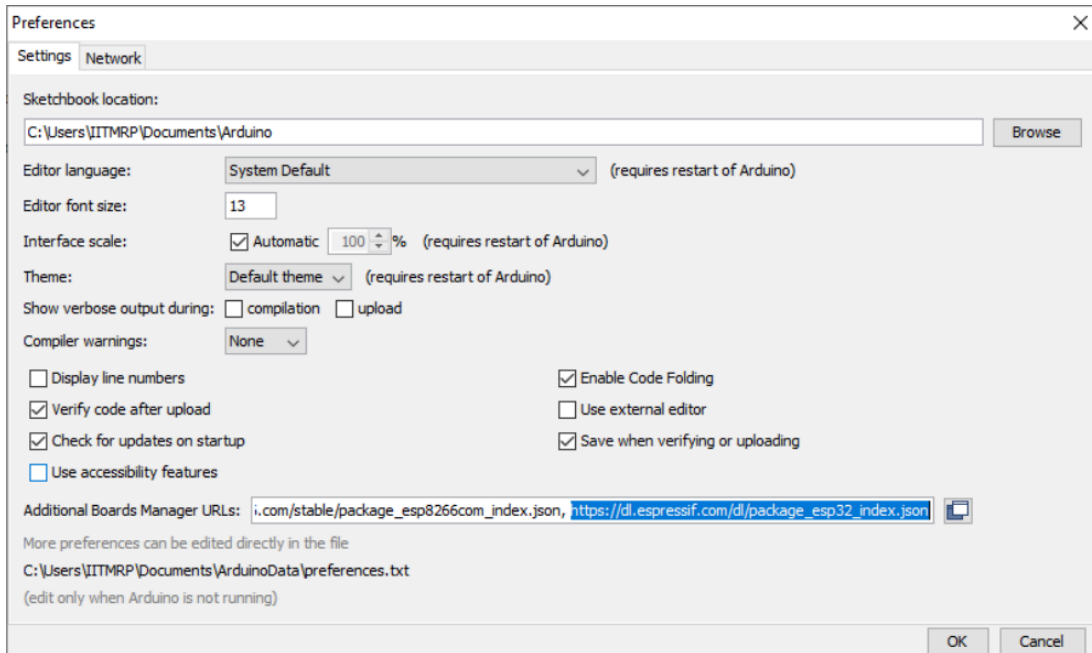
For ESP32 development:

1. Download and install the latest version of Arduino IDE for your system.
<https://www.arduino.cc/en/software>
2. From this link ,<https://www.silabs.com/developer-tools/usb-to-uart-bridge-vcp-drivers> , download the CP210x Driver for your OS (eg. CP210x Universal Windows Driver). Extract the ZIP folder. Right-click the silabser.inf file inside and select install. Follow the prompts until installation is successful. Restart your PC once installation completes.



3. Open the Arduino IDE if it's not already open. Go to File > Preferences. In 'Additional Boards Manager URLs', if the textbox is empty, paste the following links separated by a comma:
https://dl.espressif.com/dl/package_esp32_index.json
https://raw.githubusercontent.com/espressif/arduino-esp32/gh-pages/package_esp32_index.json

If the textbox already has some other links, put a comma after them and paste the above link.



4. Next, go to Tools > Board > Boards Manager. In the search bar on the top, search 'esp32' (version 2.0.11) and install the following packages. This will take some time to install.
 - a. esp32 by Espressif Systems
 - b. Arduino ESP32 Boards by Arduino

Setup for software projects

1. Install and set up Python version 3.11.8

<https://www.python.org/downloads/release/python-3118/>

DO NOT FORGET TO SELECT “ADD TO PATH” option.



2. Download and install the latest version of VSCode for your system.

<https://code.visualstudio.com/download>

<https://drive.google.com/drive/folders/18ZlhMpMPicuTx3eQF4ZH96OOGq3s5YCCQ>

Refer to this PDF to install VSCode properly.

3. Once VSCode is set up, open a new folder by going to File>New Folder.
4. Once in the new folder, press CTRL + P, then type
Python: Create Environment
This will create a virtual environment to run python in that folder.
5. In the terminal window at the bottom of screen, type
pip install opencv-python
This will install opencv for image processing.
6. Install Node Version manager. This is required to run the E-MarketPlace website.
<https://www.freecodecamp.org/news/node-version-manager-nvm-install-guide/>
Refer to this blog to properly install NVM for your system.
Once installed, go to your terminal and type
Nvm on
to activate node version manager.

Then type

```
Nvm install 20.9.0
```

to install node version 20.9.0 for your project.